UnBake: PBR Material Estimation for 3D Meshes with Baked-In Lighting using Generative Priors

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Abstract

We propose a training-free method to recover PBR materials from 3D meshes with baked-in lighting. We guide the 3D material prior of TRELLIS.2 with a physically based rebaking loss and confidence-weighted 2D intrinsic priors from RGBX.

1 Introduction

Scanned and generated 3D assets often have lighting baked into their textures, which prevents them from being correctly relit under novel illumination. Yet robotics simulation, games and content creation, and interior design all require relightable assets with PBR materials. Existing PBR texturing methods largely ignore the input texture and generate new materials from scratch, while 2D intrinsic decomposition models operate per view, producing inconsistent estimates that often retain baked shadows in the albedo. 3D generative models such as TRELLIS.2 [4] provide a strong prior over materials, but discard the existing texture entirely. Our contributions are as follows:

- **Dataset.** We curate PBR assets from Objaverse-XL [2] and ABO [1] and pair them with baked-lighting counterparts.
- **Method.** We propose a training-free approach that recovers PBR materials by combining 2D intrinsic priors from RGBX [5] with the 3D material prior of TRELLIS.2 [4].

2 Related Work

Recent generative models provide strong priors for material estimation: RGBX [5] predicts intrinsic maps such as albedo, roughness and metallic from single images, while TRELLIS.2 [4] generates PBR materials natively in a sparse 3D latent conditioned on geometry and an image. Material Anything [3] estimates materials for objects with baked lighting view by view using a trained image diffusion model. We instead combine both priors training-free, guiding TRELLIS.2 with a rebake loss under estimated lighting and fused RGBX estimates, and keep the input UV layout.

3 Data

We curate assets from Objaverse-XL [2] and ABO [1], keeping those with non-constant albedo, roughness, and metallic maps and clean, non-overlapping UVs. We then render each asset with a path tracer under randomly rotated HDR environment maps and bake the diffuse lighting, including cast shadows and ambient occlusion, into its original UV layout. Since the geometry is unchanged, this yields pixel-aligned PBR $\Leftrightarrow$ baked-lighting pairs.

Preprint.

4 Method

We first encode the input mesh into O-Voxel space and obtain its shape latent with the frozen TRELLIS.2 [4] shape encoder. For every active voxel v_i, we sample the observed baked color $\mathbf{c}_i$ and normals $\mathbf{n}_i$.

2D intrinsic priors. Before sampling, we render 16–32 views of the baked asset under neutral lighting and apply RGBX [5] to each view to obtain albedo, roughness, and metallic maps. We project every voxel into each view in which it is visible, sample the predicted values, and robustly average them across views to obtain the fused estimates $\hat{\mathbf{a}}_i$, $\hat{r}_i$, and $\hat{m}_i$. For every channel c, we compute the per-voxel standard deviation σ_i^c across views and turn it into a confidence weight

$$w_i^c = \frac{1}{1 + (\sigma_i^c / \tau^c)^2}$$

where τ^c is the median of σ^c over all voxels. Voxels seen by fewer than two views receive zero weight.

Guided sampling. We run the TRELLIS.2 [4] material stage, conditioned on the geometry and a render of the baked asset as the reference image. At selected steps t of the rectified-flow sampler, we estimate the clean latent $\hat{x}_0 = x_t - t\,\hat{v}_\theta(x_t, t)$ and decode it with the frozen material decoder $\mathcal{D}$ into per-voxel albedo $\mathbf{a}_i$, roughness r_i, and metallic m_i. Given these materials and the observed colors, we fit second-order Spherical Harmonics (SH) coefficients of the environment lighting by linear least squares, using a Lambertian model with per-voxel visibility precomputed from the geometry to account for baked shadows. Rebaking the decoded materials under this lighting yields the predicted colors $\hat{\mathbf{c}}_i$ and the rebake loss. We additionally compare the decoded materials to the fused 2D estimates:

$$\mathcal{L}_{\text{rebake}} = \frac{1}{N}\sum_{i=1}^{N} \text{Huber}(\hat{\mathbf{c}}_i - \mathbf{c}_i) \qquad \mathcal{L}_{\text{2D}} = \sum_{c \in \{a,r,m\}} \lambda_c \frac{\sum_i w_i^c \left| p_i^c - \hat{p}_i^c \right|}{\sum_i w_i^c}$$

where p_i^c is the decoded value of channel c and $\hat{p}_i^c$ the corresponding fused estimate. We then correct the clean-latent estimate by backpropagating the total loss $\mathcal{L} = \mathcal{L}_{\text{rebake}} + \beta\,\mathcal{L}_{\text{2D}}$ through $\mathcal{D}$ only and continue sampling from the corrected estimate. All network weights remain frozen.

Output. Finally, we bake the decoded per-voxel materials into the mesh's original UV layout by sampling the voxel grid at each texel. This ensures the exact same layout as the users input.

5 Evaluation

We report per-channel MAE and PSNR against ground-truth materials, and MAE, PSNR, SSIM, and LPIPS on renders relit under novel lighting. We compare against an input-as-albedo baseline, unguided TRELLIS.2, and Material Anything [3].

6 Milestones

Week	Goals
1 (10/26)	Curate dataset and bake lighting; run TRELLIS.2 and RGBX; evaluation pipeline.
2 (11/02)	Unguided TRELLIS.2 and Material Anything baselines; voxel-to-view projection.
3 (11/09)	Rebake function, visibility computing, and SH fitting; RGBX view fusion and confidence weights.
4 (11/16)	Guided sampling with rebake loss.
5 (11/23)	Add 2D prior loss; compare against rebake-only.
6 (11/30)	Buffer for bug fixes and final evaluation.
7 (12/07)	Final write-up.

References

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